KISII UNIVERSITY

School of Information Science and Technology
Department of Computing Sciences

FOURTH-YEAR UNDERGRADUATE PROJECT
PROJECT PROPOSAL GUIDELINES

Purpose. The project proposal is a formal plan of the proposed undergraduate project. It should demonstrate that the student has identified a genuine and clearly defined problem, reviewed relevant existing work, identified a gap or need, and developed a realistic plan for designing, developing, and evaluating a solution.

1. 1. Project Title Selection and Approval

The project title should clearly communicate **what is being developed or investigated, the problem/domain being addressed, and the intended context or users where appropriate**.

A good title should be:

- specific, concise, meaningful, and technically appropriate;
- based on a clearly identifiable problem or need;
- achievable within the available time, skills, and resources;
- consistent with the proposed objectives and scope; and
- neither too broad nor unnecessarily complicated.

Avoid vague titles such as *"A Management System"* or titles containing features that the project will not realistically implement.

Important: Every project title must be submitted to and approved by the supervisor before the student proceeds to develop the proposal. Students should not substantially change an approved title without consulting the supervisor.

2. 2. Recommended Proposal Structure

The proposal should normally contain the following:

1. **Preliminary Pages:** Title page, declaration (where required), abstract, table of contents, and lists of figures/tables where applicable.
2. **Chapter One – Introduction:** Background of the study, problem statement, main objective, specific objectives, research questions (where applicable), significance, scope, limitations, and definition of key terms.
3. **Chapter Two – Literature Review:** Relevant concepts/theories, review of related studies and/or existing systems, comparison of existing approaches, and identification of the research/knowledge or practical gap addressed by the proposed project.
4. **Chapter Three – Methodology and Proposed Design:** Development/research methodology and its justification, requirements analysis, proposed system architecture/design, technologies and tools, data collection methods where applicable, and proposed testing/evaluation approach.
5. **References:** Complete list of all sources cited in the proposal.
6. **Appendices:** Supporting material where necessary, such as questionnaires, interview guides, work plans, or additional diagrams.

Note: The exact structure may be adjusted according to the nature of the project and the supervisor's guidance.

3. 3. Writing and Content Requirements

Students are expected to:

- write clearly, logically, and in formal academic language;
- ensure that the problem statement, objectives, methodology, and proposed solution are logically aligned;
- make objectives specific and measurable, using action verbs such as *analyse, design, develop, implement, integrate, evaluate,* and *test*;
- provide evidence from relevant academic and authoritative sources;
- explain and justify important methodological and technological choices;
- use appropriate diagrams, tables, models, and figures where they improve clarity; and
- avoid unnecessary material included merely to increase the page count.

The proposal should normally be **15–30 pages**, excluding preliminary pages, references, and appendices, unless otherwise directed by the supervisor or department.

4. 4. Literature Review

The literature review should demonstrate understanding rather than merely list sources. Students should compare relevant studies, systems, methods, or technologies and identify their strengths, limitations, and relevance to the proposed project.

The review should ultimately answer:

What has already been done, what is missing or inadequate, and how does this project address the identified gap?

Priority should be given to peer-reviewed journal articles, conference papers, academic books, theses, official technical doc-

umentation, government publications, and other credible sources. Websites and informal sources should not form the main evidence for academic arguments.

5. 5. Referencing and Citation

Students may use either of the following referencing styles:

IEEE	Recommended for Computing and Information Technology; uses numbered citations such as [1], [2], [3].
APA	An acceptable alternative; uses author–date citations such as (Otieno, 2025).

Choose ONE style and use it consistently throughout the entire proposal. Do not mix IEEE and APA conventions.

Students must cite sources when using another person's:

- ideas, definitions, theories, findings, statistics, or arguments;
- algorithms, models, technical approaches, or other intellectual contributions; or
- figures, tables, diagrams, or material reproduced/adapted from another source.

Every source cited in the text must appear in the reference list, and every source in the reference list should be cited in the text. Students are encouraged to use a reference manager such as Zotero, Mendeley, or similar software, provided the generated references are checked for accuracy.

6. 6. Academic Integrity and Plagiarism

Students must submit original academic work. Copying material from websites, previous projects, classmates, journal articles, or other sources without appropriate acknowledgement constitutes plagiarism.

Paraphrasing requires more than changing a few words: students should understand the source, express the idea in their own words, and provide the appropriate citation. Direct quotations, where necessary, must be clearly identified and cited.

Students are responsible for verifying all references, facts, technical claims, and content included in their proposals. The use of AI-assisted tools must comply with current university policies on academic integrity and responsible use of such tools.

7. 7. Formatting and Presentation

Unless the department provides a separate official template, use the following minimum formatting requirements:

- **Paper:** A4; **Font:** Times New Roman, 12 pt; **Line spacing:** 1.5.
- **Margins:** approximately 1 inch on all sides.
- Number chapters, sections, tables, and figures consistently.
- Every table and figure should have a caption and should be discussed or referenced in the accompanying text.
- Use consistent page numbering, headings, terminology, and referencing style.

8. 8. Submission and Approval Process

The following sequence should be followed:

Step	Required Activity
1	Identify a relevant problem/domain and propose a suitable project title.
2	Submit the proposed title to the supervisor for review and approval.
3	After title approval, develop the proposal in consultation with the supervisor.
4	Submit drafts for supervisor review and incorporate the required corrections.
5	Complete the proposal and present/defend it where required.

Students should not proceed to full proposal development on an unapproved title. Approval of a title does not automatically imply approval of the final proposal; the proposal must demonstrate a sound problem, objectives, literature review, methodology, and feasible project plan.

9. 9. Responsible Use of AI-Assisted Tools

Students may use AI-assisted tools (e.g., ChatGPT, Gemini, Claude, Copilot) as **support tools** for brainstorming, clarifying concepts, improving grammar, or identifying areas for further research. **AI-generated content must not be copied and submitted as original work.** The student remains fully responsible for the originality, accuracy, and academic quality of the submitted work.

Students must:

- critically review and verify all AI-generated content before use;
- rewrite and develop ideas in their own words and understanding;
- independently verify all references and sources suggested by AI tools;

- never submit fabricated or unverifiable citations;
- not use AI as a substitute for their own analysis or project work; and
- comply with University policies on academic integrity and AI use.

Note: AI output is not a credible academic source. Cite the **original sources** behind any claim, not the AI tool itself. Supervisors may require students to explain or defend any part of their submitted work.

Key principle: A good project proposal is not judged by its length, but by the clarity of the problem, quality of the literature, justification of the proposed work, and feasibility of the methodology.